

Question for a Uni Design Project: Is the massive energy footprint of AI actually on your radar?

Posted by u/TalNix77 • • 0 points • 6 comments

Hi everyone,

I'm a design student researching the "invisible" energy consumption of AI for a university project.

While the utility of tools like ChatGPT is obvious, the physical resources required to run them are massive. Studies suggest that a single generative AI query can consume significantly more energy than a standard web search (some estimates range from 10x to 25x more).

I'm looking for honest perspectives on this:

1. **Awareness:** Before reading this, were you actually aware of the scale of energy difference between a standard search and an AI prompt? Or is that completely "invisible" in your daily usage?
2. **Impact on Usage:** Does the energy intensity play any role in how you use these tools? Or is the utility simply the only factor that matters for your workflow?
3. **Value vs. Waste:** Do you view this high energy consumption as a fair investment for the results you get, or does the current technology feel inefficient to you?

I'm trying to get a realistic picture of whether this topic actually plays a role in users' minds or if performance is the priority.

Comments

u/Energy_Balance • 1 points • 2025-12-07 02:40 UTC

There have been many reports on AI data center financing, debt, the business models of the AI companies, and the probability of bankruptcies.

As you surmise, there are 2 parts: training and response to queries. Training has a huge cost, and

there are plans for 2-4GW single data centers, but we might only need 10-20 of those in the US. The training code can stop and start on demand. So training is an ideal flexible load - run it when the grid is lightly loaded and energy is cheap on the wholesale market. Then stop training from 4PM-10PM when the load is high, or during your hottest and coolest weather hours and days.

Some of the AI companies have revenues, it's not a free service for those companies. It would be good UX to return to the customer the energy cost per query, with the amortized training energy cost as a separate item. Then they could break out the capital cost, and staffing cost too.

u/gonyere • 1 points • 2025-12-06 12:43 UTC

Yes. It's a huge part of why I don't use AI. It's just not worth the price - it may be "free" to me. But it's incredibly wasteful for the world at large.

u/carboncritic • 3 points • 2025-12-06 04:14 UTC

1. Yes

2. Yes

3. No

u/Cominginbladey • 3 points • 2025-12-06 03:01 UTC

I question the premise that the "utility of ChatGPT is obvious.

AI is a speculative investment bubble fueled by excess capital. The only question is who gets left holding the bag. The electric industry is using it as a justification for over-building. Soon poor people will not be able to afford electricity.

We are burning the forests, draining our water, impoverishing people and shovelling our money into tech bros' pockets so we can make videos of a cat doing the moonwalk.

u/glyptometa • 1 points • 2025-12-06 01:07 UTC

Awareness - very much so, yes. It's a significant risk in the context of global heating. It would be hard to miss if you have any interest in the risks of rapid climate change.

Impact on usage is hard to answer. If you use critical thinking to find answers, you need to drill down to reliable information either way.

Value vs. Waste - I see it as wasteful and unnecessary. Just a different way to search, with no true time savings, because of needing to drill down anyway. Plus it's wrong 15% or 20% of the time anyway.

Before enshittification, you could go to three links and if two out of three said the same thing, you probably had what you needed. Now you just need to review more sources. I use 4 out of 5 or a reputable journal article.

u/michaelhoney • 1 points • 2025-12-06 01:06 UTC

I'm very aware of AI energy consumption, but I don't know what the solid numbers are. For example, I just read at <https://www.thegreattransition.org> that AI will only account for 3% of global energy use in 2040. If so, that sounds like a good deal – but is that number realistic?

Happy to discuss this more if you want

